

Solutions — Lesson 01

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These are full worked solutions for every exercise in `01-pen-tracking.md`. Try the exercises before peeking; the value of an exercise is in attempting it, even unsuccessfully.

Inline exercises

Exercise 1.1 — Body v , ω , ICR for $v_l = 8$, $v_r = 12$ cm/s.

Given $v_l = 8$ cm/s, $v_r = 12$ cm/s, $W = 20.4$ cm.

(a) Linear speed:

$$v = \frac{1}{2}(v_l + v_r) = \frac{1}{2}(8 + 12) = 10 \text{ cm/s.}$$

(b) Angular speed:

$$\omega = \frac{v_r - v_l}{W} = \frac{12 - 8}{20.4} = \frac{4}{20.4} \approx 0.196 \text{ rad/s.}$$

(c) ICR distance from M :

$$R = \frac{v}{\omega} = \frac{10}{0.196} \approx 51 \text{ cm.}$$

Sign of ω is positive, which by convention means CCW rotation. Looking from above with x forward and y to the left, CCW rotation about M is a left turn. So **the body is turning left**, on a circle of radius ≈ 51 cm centred to the left of M .

Sanity check: the right wheel is moving faster than the left wheel, so the body should yaw counter-clockwise (left). ✓

Exercise 1.2 — Compute $\det J(\theta)$.

Starting from

$$J(\theta) = \begin{pmatrix} \cos \theta & -(p_x \sin \theta + p_y \cos \theta) \\ \sin \theta & p_x \cos \theta - p_y \sin \theta \end{pmatrix},$$

apply the 2×2 determinant formula $\det = ad - bc$:

$$\det J = \cos \theta (p_x \cos \theta - p_y \sin \theta) - \sin \theta (-(p_x \sin \theta + p_y \cos \theta)).$$

Expand:

$$= p_x \cos^2 \theta - p_y \sin \theta \cos \theta + p_x \sin^2 \theta + p_y \sin \theta \cos \theta.$$

The $p_y \sin \theta \cos \theta$ terms cancel, and $\cos^2 \theta + \sin^2 \theta = 1$, so:

$\det J(\theta) = p_x$

Independent of θ and of p_y .

Sanity check: in the special case $\theta = 0$, $J = \begin{pmatrix} 1 & -p_y \\ 0 & p_x \end{pmatrix}$, an upper triangular matrix whose determinant is the product of the diagonal: $1 \cdot p_x = p_x$. ✓

Exercise 1.3 — Pen at $(0, 10.2)$: is `trace` available?

The pen sits on the wheel-axis line ($p_x = 0$), specifically directly on top of the left wheel. By the result of Exercise 1.2, $\det J = p_x = 0$, so J is singular at every heading and J^{-1} does not exist. The `trace` primitive depends on inverting J at every timestep, so **it cannot work for this pen position**.

The codebase explicitly rejects `trace` for $p_x = 0$ (see `_plan_trace` in `drawingrobot/script.py`). The alternative is `line_to`, which plans rotate-translate-rotate setup + forward and does not invert J — and in this special case the setup translation $\Delta = (R(\theta_{\text{cur}}) - R(\theta_{\text{new}}))(p_x, p_y)^\top$ has the wheel-axis-aligned p_y component preserved, so the corner curvature is concentrated at the vertex with magnitude $|p_y(1 - \cos \Delta\theta)|$ for the 90° case, $\approx |p_y|$.

Exercise 1.4 — Single-arc bound for $(p_x, p_y) = (14.4, 0)$.

- (a) The single-arc lemma gives $r_{\text{pen}} \geq |p_x| = 14.4$ cm. Saturated when the ICR is placed at $R = p_y = 0$, i.e. directly under M — but that is exactly in-place rotation, which gives $r_{\text{pen}} = |p_{\text{body}}| = \sqrt{14.4^2 + 0^2} = 14.4$ cm. So **14.4 cm**.
- (b) `goto` plans a rotate-then-forward at each leg. The rotation is in-place, so its pen radius is $|p_{\text{body}}| = 14.4$ cm. **14.4 cm at every corner**.

- (c) The square's edge length is 30 cm but each corner arc has radius 14.4 cm — almost half the edge. The arcs span a substantial fraction of each leg, so the pen path between corners is dominated by a large fillet rather than a straight segment. The user sees a smooth blob rather than a square. (`line_to` and `trace` are designed precisely to fix this — `line_to` by localising the corner curvature at the vertex, and `trace` by sidestepping the single-arc lemma entirely.)

Sanity check: the simulator screenshot in §5 (Figure 1.7, left panel) shows exactly this — `goto` produces a chaotic shape because the corner arcs span the legs. ✓

Exercise 1.5 — (v, ω) for $\dot{P}_{\text{des}} = (0, 10)$ at $\theta = 0$, $p = (14.4, 0)$.

Pen offset $(p_x, p_y) = (14.4, 0)$, heading $\theta = 0$, so $\sin \theta = 0$, $\cos \theta = 1$, and

$$a = p_x \sin \theta + p_y \cos \theta = 0, \quad b = p_x \cos \theta - p_y \sin \theta = 14.4.$$

Apply the inverse-Jacobian formula with $\dot{P}_{\text{des}} = (\dot{P}_x, \dot{P}_y) = (0, 10)$:

$$v = \frac{b \dot{P}_x + a \dot{P}_y}{p_x} = \frac{14.4 \cdot 0 + 0 \cdot 10}{14.4} = 0,$$

$$\omega = \frac{-\sin \theta \dot{P}_x + \cos \theta \dot{P}_y}{p_x} = \frac{0 \cdot 0 + 1 \cdot 10}{14.4} = \frac{10}{14.4} \approx 0.694 \text{ rad/s}.$$

So $(v, \omega) = (0, 0.694)$ rad/s. **The body spins in place.**

That looks counter-intuitive — the pen is supposed to move sideways in the world, why is the body not translating? The answer: with $\theta = 0$ the body's heading is along world $+x$, and a pen at $(p_x, p_y) = (14.4, 0)$ sits in front of M along that heading. The non-holonomic constraint forbids M from moving in world $+y$ without a rotation. The only way the pen-tip moves in world $+y$ at this instant is by the body rotating about M — which swings the pen-tip perpendicular to $\overrightarrow{MP}$, i.e. exactly in the world $+y$ direction at this configuration.

Sanity check: a pure rotation $\omega = 10/14.4$ rad/s about M moves a point at distance 14.4 from M at tangential speed $\omega \cdot 14.4 = 10$ cm/s perpendicular to $\overrightarrow{MP}$. Since $\overrightarrow{MP}$ is along $+x$, the tangential direction is $+y$. ✓

Exercise 1.6 — **Pen at (0.5, 7): trace available, but...**

- (a) Yes, $p_x = 0.5 \neq 0$, so $\det J = 0.5 \neq 0$, J is invertible, `trace` is available.
- (b) The undesirable issue is **gain**. The inverse Jacobian has $1/p_x$ as a prefactor: with $p_x = 0.5$ cm, even moderate commanded pen velocities produce large ω — for $\dot{P}_{\text{des}} = 10$ cm/s in the unfavourable direction, $\omega = O(\dot{P}/p_x) = O(20)$ rad/s, which is well above the simulator's default π rad/s and certainly above any plausible physical wheel speed. The wheels saturate, the inverse Jacobian's request is not

met, and the pen drifts off the polyline. In short: **trace is well-defined but the actuator authority is wrong** — the controller asks for body rotations the wheels cannot deliver.

End-of-chapter

Exercise 1.E1 — Diff-drive expressions and limits.

For wheel speeds v_l, v_r on a chassis of width W :

$$v = \frac{1}{2}(v_l + v_r), \quad \omega = \frac{v_r - v_l}{W}, \quad R = \frac{v}{\omega} = \frac{W}{2} \frac{v_l + v_r}{v_r - v_l}.$$

Limits:

- $v_l = v_r = v_0 \Rightarrow \omega = 0, v = v_0, R = \infty$. Body translates straight along its heading at speed v_0 . ✓
 - $v_l = -v_r = -v_0 \Rightarrow v = 0, \omega = 2v_0/W, R = 0$. Body spins in place about M at angular speed $2v_0/W$. ✓
 - $v_l = 0, v_r = v_0 \Rightarrow v = v_0/2, \omega = v_0/W, R = W/2$. Body pivots about the *left* wheel (which is sitting still). The radius $R = W/2$ confirms the ICR sits exactly at the left wheel position. ✓
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Exercise 1.E2 — Pen position at world pose $(5, 2, \pi/4)$, body offset $(10, -3)$.

Apply

$$P_x = x + p_x \cos \theta - p_y \sin \theta, \quad P_y = y + p_x \sin \theta + p_y \cos \theta.$$

With $\theta = \pi/4$, $\cos \theta = \sin \theta = 1/\sqrt{2} \approx 0.7071$:

$$P_x = 5 + 10 \cdot 0.7071 - (-3) \cdot 0.7071 = 5 + 7.071 + 2.121 = 14.19 \text{ cm.}$$

$$P_y = 2 + 10 \cdot 0.7071 + (-3) \cdot 0.7071 = 2 + 7.071 - 2.121 = 6.95 \text{ cm.}$$

So $\boxed{P \approx (14.19, 6.95) \text{ cm}}$.

Sanity check: the pen offset $(10, -3)$ has magnitude $\sqrt{100 + 9} \approx 10.44$ cm; it should land at distance 10.44 from $M = (5, 2)$. Computing: $\sqrt{(14.19 - 5)^2 + (6.95 - 2)^2} = \sqrt{84.5 + 24.5} = \sqrt{109.0} \approx 10.44$ cm. ✓

Exercise 1.E3 — No non-holonomic constraint on P when $p_x \neq 0$.

The body constraint reads $\dot{x} \sin \theta - \dot{y} \cos \theta = 0$ at every instant: the wheel midpoint M has zero velocity in the body $\hat{y}_b$ -direction.

Suppose, for contradiction, the pen P obeys an analogous constraint of the form

$$\dot{P}_x f(\theta) + \dot{P}_y g(\theta) = 0$$

for some non-trivial (f, g) . Then the row vector (f, g) must annihilate $J(\theta)(v, \omega)^\top$ for every admissible (v, ω) . So (f, g) must be in the *left null space* of J . But $\det J = p_x$ — and if $p_x \neq 0$, J has full rank, so its left null space is $\{0\}$. The only (f, g) that works is the zero vector, i.e. there is **no constraint**.

Physical interpretation: when the pen sits off the wheel-axis line, M 's sideways immobility is partly compensated by the rotation of the body about M , which translates the pen sideways from M 's heading. The pen velocity gets a 2-D worth of authority — a translation contribution from v along $\hat{x}_b$ and a rotation contribution from ω perpendicular to $\overrightarrow{MP}$ — and these two basis vectors are linearly independent precisely when $\overrightarrow{MP}$ has a non-zero body $\hat{x}_b$ -component (i.e. $p_x \neq 0$).

When $p_x = 0$ the rotation contribution becomes parallel to $\hat{y}_b$ — both basis vectors collapse to a 1-D span, and the pen recovers a non-holonomic constraint (its velocity must lie along $\hat{y}_b$ from M 's perspective).

Exercise 1.E4 — Verify $J J^{-1} = I_2$.

Let $a = p_x \sin \theta + p_y \cos \theta$, $b = p_x \cos \theta - p_y \sin \theta$. Then

$$J = \begin{pmatrix} \cos \theta & -a \\ \sin \theta & b \end{pmatrix}, \quad J^{-1} \stackrel{?}{=} \frac{1}{p_x} \begin{pmatrix} b & a \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

Compute $J J^{-1}$ entry by entry. The $(1, 1)$ entry:

$$\frac{1}{p_x} [\cos \theta \cdot b + (-a) \cdot (-\sin \theta)] = \frac{1}{p_x} [b \cos \theta + a \sin \theta].$$

Substituting,

$$\begin{aligned} b \cos \theta + a \sin \theta &= (p_x \cos \theta - p_y \sin \theta) \cos \theta + (p_x \sin \theta + p_y \cos \theta) \sin \theta \\ &= p_x (\cos^2 \theta + \sin^2 \theta) + p_y (-\sin \theta \cos \theta + \cos \theta \sin \theta) = p_x. \end{aligned}$$

So the $(1, 1)$ entry is $p_x/p_x = 1$. ✓

The $(1, 2)$ entry:

$$\frac{1}{p_x} [\cos \theta \cdot a + (-a) \cdot \cos \theta] = 0. \checkmark$$

The $(2, 1)$ entry:

$$\frac{1}{p_x} [\sin \theta \cdot b + b \cdot (-\sin \theta)] = 0. \checkmark$$

The $(2, 2)$ entry:

$$\frac{1}{p_x} [\sin \theta \cdot a + b \cdot \cos \theta] = \frac{1}{p_x} [a \sin \theta + b \cos \theta] = \frac{p_x}{p_x} = 1. \checkmark$$

So $J J^{-1} = I_2$, confirming the formula.

Exercise 1.E5 — Saturation at a sharp corner, $p_x = 14.4$, $v_{\text{pen}} = 12$, $\Delta t = 1/120$.

- (a) Heading change required so that the pen-tip can pivot from moving in $+x$ to moving in $+y$. With $\theta_0 = 0$ and pen at body offset $(14.4, 0)$, the pen velocity column 1 of J is $(\cos \theta, \sin \theta) = (1, 0)$ at $\theta = 0$ — perfect for $\dot{P} = (12, 0)$. After the corner we want $\dot{P} = (0, 12)$, which is achieved at $\theta_1 = \pi/2$ (heading along $+y$). So the body must change heading by $|\Delta\theta| = \pi/2 \approx 1.571$ rad.
- (b) Spread over a single timestep $\Delta t = 1/120$ s, this requires

$$|\omega|_{\text{peak}} \approx \frac{|\Delta\theta|}{\Delta t} = \frac{\pi/2}{1/120} = 60\pi \approx 188.5 \text{ rad/s.}$$

- (c) The simulator’s default in-place rotation speed is $\omega_{\text{rot}} = \pi$ rad/s, which corresponds to wheel-tip speeds $v_l = -\omega W/2 = -\pi \cdot 10.2 \approx -32$ cm/s on each side. The peak $|\omega|$ at the corner exceeds this by a factor of 60, so **yes, the wheels saturate severely** at the corner.

What actually happens in the simulator: the per-timestep wheel-velocity demand is enormous, and (depending on whether the simulator clips wheel speeds) either (i) the pose-integration step takes a single huge $\omega\Delta t$ that swings the body too much, or (ii) the wheels saturate, the demanded $\Delta\theta$ is not delivered, and the pen “rounds” the corner with a small fillet of radius proportional to the missing rotation. In practice we don’t notice this in the trace plots because the integration is done with the formulas as given — but if the codebase later adds a wheel-speed clip, this corner will be the first place the trace primitive visibly degrades.

The right cure is in §4’s “more robust planner” sketch: bound $|\omega|$ to a ceiling and round corners only when it would otherwise saturate.

Exercise 1.E6 — Two arcs cannot beat $|p_x|$.

Each arc command, by the single-arc lemma of §3, paints a pen circle of radius $\geq |p_x|$ — the bound is on the *instantaneous curvature radius* of the pen during that command. So during the first command, the pen’s curvature radius is $\geq |p_x|$; during the second command, also $\geq |p_x|$. The transition between the two commands is instantaneous (the command boundary), so the pen’s pose at the join is just whatever the first command produced — there is no instantaneous “kink” allowed in the pen’s curvature, only a discontinuity in the curvature value.

So the pen’s radius of curvature is bounded below by $|p_x|$ at every instant during the two-arc composition. The smallest pen circle inscribed in such a path has radius $\geq |p_x|$. The bound is unchanged.

By induction, the same holds for any **finite** composition of constant-input commands: the worst-case pen curvature radius across the whole composition is $\geq |p_x|$.

What `trace` does differently: it does not issue constant-input commands. It updates (v, ω) at every timestep, and at the corner specifically it asks for instantaneous curvature radius approaching zero. The pen's *average* curvature radius over a timestep is still finite (limited by $v_{\text{pen}}\Delta t$), but the bound $r \geq |p_x|$ no longer applies because the inputs are no longer constant. Many small commands per edge — at $\Delta t = 1/120$ s, around 360 commands per 30 cm leg — give the pen velocity-discontinuity authority that the single-arc lemma denied.